

Persistence of intact HIV-1 proviruses in the brain during suppressive antiretroviral therapy

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Abstract

HIV-1 reservoir cells that circulate in peripheral blood during suppressive antiretroviral therapy (ART) have been well characterized, but little is known about the dissemination of HIV-1-infected cells across multiple anatomical tissues, especially the central nervous system (CNS). Here, we performed single-genome, near full-length HIV-1 next-generation sequencing to evaluate the proviral landscape in distinct anatomical compartments, including multiple CNS tissues, from 3 ART-treated participants at autopsy. While lymph nodes and, to a lesser extent, gastrointestinal and genitourinary tissues represented tissue hotspots for the persistence of intact proviruses, we also observed intact proviruses in CNS tissue sections, particularly in the basal ganglia. Multi-compartment dissemination of clonal intact and defective proviral sequences occurred across multiple anatomical tissues, including the CNS, and evidence for the clonal proliferation of HIV-1-infected cells was found in the basal ganglia, in the frontal lobe, in the thalamus and in periventricular white matter. Deep analysis of HIV-1 reservoirs in distinct tissues will be informative for advancing HIV-1 cure strategies.

eLife assessment

This **important** study uses near full-length HIV-1 sequencing to examine proviral persistence in various tissues derived from three individuals who received antiretroviral therapy until time of death. Intact as well as defective HIV-1 proviruses are found at various anatomical sites including the central nervous system, results that are **convincing** and relevant for our understanding of latent viral reservoirs, especially in the brain.

Introduction

Combination antiretroviral therapy has extended the life expectancy of people living with HIV (PLWH) to near-normal levels, but it is clear that antiretroviral therapy (ART), in its present form, does not lead to a sustained, drug-free remission of HIV-1 infection; instead, a long-lived reservoir

of virally-infected cells persists despite ART (1, 2) and can effectively fuel rebound viremia in case of treatment interruptions. Hence, antiretroviral therapy needs to be taken life-long; finding a way to eliminate HIV-1 reservoir cells remains critical to finding a cure for HIV-1 infection.

HIV-1 reservoir cells that circulate in the peripheral blood have been well-described in recent years (3). These circulating reservoir cells mostly consist of memory CD4 T cells that persist life-long and harbor chromosomally-integrated viral DNA, also referred to as a “provirus”. The pool size of HIV-1-infected cells during suppressive antiretroviral therapy is frequently maintained or replenished by clonal proliferation of viral reservoir cells, during which an identical viral DNA copy is passed on to daughter cells (4–7). Important advances in recent technology development, including single-genome near full-length proviral sequencing, have demonstrated that the vast majority of HIV-1 DNA species encountered in ART-treated persons are defective (8–10), mostly due to errors occurring during reverse transcription of viral RNA into DNA. Such errors can lead to large deletions, premature stop codons, or defects in the viral packaging signal region; moreover, specific host proteins such as APOBEC-3G/3F can induce lethal hypermutations into the proviral sequence. The ability of viral reservoir cells to persist indefinitely is frequently attributed to very limited or absent proviral gene transcription; this viral latency protects infected cells from host immune responses and reduces possible cytopathic effects associated with viral gene expression. However, recent studies have emphasized that proviral gene expression can be ongoing in some HIV-1-infected cells during antiretroviral therapy (11), typically when proviruses are integrated in immediate proximity to activating chromatin marks (12).

Much less is currently known about HIV-1 reservoir cells in other body compartments that are more difficult to access for analytic purposes. Lymph nodes and lymphoid tissues in the gastrointestinal tract harbor the vast majority of all lymphocytes and are likely to represent a prime location for the persistence of virally-infected cells (13–16); however, studies that interrogate viral reservoir cells in these tissues remain limited. Even less is known about HIV-1 persistence in other organs, and, in particular, in the CNS, although recent studies have started to explore viral reservoir cells in such specific body compartments (17). In the present study, we used single-genome proviral sequencing to conduct a detailed analysis of HIV-1 proviral sequences in autopsy samples from up to n=18 different organ systems from three study participants.

Methods

Study Participants

HIV-1-infected study participants were recruited at the Massachusetts General Hospital (MGH) and the Brigham and Women’s Hospital in Boston, MA. Fresh tissues were sampled during routine autopsy according to protocols approved by the Institutional Review Board and cryopreserved for future study according to standard protocols. The clinical characteristics of study participants are summarized in **Figure S1**.

HIV-1 DNA quantification by IPDA

Tissue samples were dissected and subjected to genomic DNA extraction using the DNeasy Blood and Tissue Kit (QIAGEN DNeasy, #69504). HIV-1 DNA was analyzed by the Intact Proviral DNA Assay (IPDA), using primers and probes described previously (18). PCR was performed using the following program: 95 °C for 10 min, 45 cycles of 94 °C for 30 s and 59 °C for 1 min, 98 °C for 10 min. The droplets were subsequently read by the QX200 droplet reader (Bio-Rad), and data were analyzed using QuantaSoft software (Bio-Rad).

Near full-length HIV proviral sequencing

Genomic DNA diluted to single HIV-1 genome levels based on Poisson distribution statistics and IPDA results was subjected to near full-genome HIV-1 amplification using a one-amplicon approach (9 [9](#), 19 [19](#)). PCR products were visualized by agarose gel electrophoresis. Amplification products were individually subjected to Illumina MiSeq sequencing at the MGH DNA Core facility. The resulting short reads were de novo assembled using Ultracycler v1.0 and aligned to HXB2 to identify large deleterious deletions (<8000 bp of the amplicon aligned to HXB2), out-of-frame indels, premature/lethal stop codons, internal inversions, or 5'-LTR defect (≥ 15 bp insertions and/or deletions relative to HXB2), using an automated in-house pipeline written in Python scripting language (<https://github.com/BWH-Lichterfeld-Lab/Intactness-Pipeline> [20](#)). The presence/absence of APOBEC-3G/3F-associated hypermutations were determined using the Los Alamos HIV Sequence Database Hypermut 2.0 program. Viral sequences that lacked all defects listed above were classified as “genome-intact”. Multiple sequence alignments were performed using MUSCLE (20 [20](#)). Phylogenetic analyses were conducted using MEGA X, applying maximum likelihood approaches. Viral sequences were considered clonal if they had completely identical consensus sequences; single-nucleotide variations in primer binding sites were excluded for clonality analysis. When viral DNA sequences were undetectable, data were reported as LOD (limit of detection), calculated as 0.5 copies per maximum number of cells tested without target identification. The sensitivity of proviral species to broadly-neutralizing antibodies (bnAb) was estimated by calculating the number of amino acid signature sites associated with sensitivity to four bnAb classes within the env amino acid sequence from each intact provirus, as previously described (21 [21](#)).

Data Analysis and Statistics

Data are summarized as bar graphs. Phylogenetic relationships were evaluated using maximum-likelihood phylogenetic trees. Images were prepared using Adobe Illustrator. Due to confidentiality concerns, proviral sequencing data cannot be publicly released, but will be made available upon reasonable request and after signing an institutional data-sharing agreement.

Results

Frequency of intact and defective proviruses in tissue compartments

To investigate the proviral landscape across multiple anatomical tissues, including the central nervous system (CNS), we focused on 3 participants from whom post-mortem tissue samples were available for HIV-1 research. Tissue samples from the CNS and other organs were collected by a rapid (<24 hour) autopsy after death. The clinical and demographic characteristics of these study participants are shown in **Figure S1A**. All study participants adhered to antiretroviral treatment until death; plasma viral loads were undetectable by commercial assays in study participants 1 and 2, in whom 14 and 15 different tissue sections were sampled, respectively (**Figure S1B**). Organ-specific tissues analyzed in these two study participants included lymph node, spleen, colon, liver, pancreas, kidney, thyroid gland, and adrenal gland; in the female study participant 1, ovarian and uterus tissues were analyzed, while in the male study participant 2, prostate and testicular tissues were studied. In both of these study participants, four different CNS tissue sections (basal ganglia, thalamus, frontal lobe, and occipital lobe) were collected for investigation. In study participant 3, plasma viral load was 136 copies/ml at the time of death; 5 different tissue sections from the CNS (basal ganglia, thalamus, occipital lobe, frontal lobe, periventricular white matter) were analyzed in this person.

Near full-length single-template next-generation HIV-1 proviral sequencing was performed to profile the proviral reservoir landscape in single-molecule resolution in tissue samples. The number of cells analyzed from each organ in each participant is listed in **Figure S1B**. In total, 846.53, 425.54, and 199.66 million cells were assayed in study participants 1, 2, and 3, respectively, resulting in 1471.73 million cells analyzed in all study participants combined. A total of 1497 individual proviral sequences were amplified, of which $n=497$ were selected for next-generation sequencing based on their amplicon size on gel electrophoresis; the remaining sequences were classified as proviruses with large deletions. All amplicons ($n=74$) from CNS tissues were sequenced, regardless of their length. Using a previously-described computational pipeline to identify lethal defects in proviral sequences, we identified 48 proviruses (3.21% of all proviruses) that met our criteria for genome-intactness (**Figure 1A-C**); this number is consistent with the small number of genome-intact proviruses detected in previous studies. Many sequences, both genome-intact and defective, were identified multiple times, consistent with clonal proliferation of infected cells (**Figure 1B**), as reported in prior work (7, 9, 10, 22).

To evaluate HIV-1 persistence in selected tissue compartments, the frequencies of total, intact, and defective HIV-1 proviruses in each tissue were analyzed (**Figure 2A-C**). Intact proviral sequences were only detected in 8 tissue sites, including basal ganglia, periventricular white matter, lymph node, spleen, colon, kidney, prostate, and thyroid (**Figure 2B**). The numbers of intact HIV-1 sequences in these 8 tissues varied from 0.01 to 0.6 copies per million cells. Consistent with previous studies, the frequency of intact proviral sequences was highest in the lymph node in participant 1 (0.52 intact proviruses/million cells) and participant 2 (0.58 intact proviruses/million cells), followed by kidney, spleen, colon, and basal ganglia in participant 1 and by prostate, spleen, thyroid in participant 2. Intact proviruses were detected in the basal ganglia in study participant 1 (frequency of 0.015/million cells) and in study participant 3 (0.030 intact proviruses/million cells). Moreover, one intact provirus was also detected in periventricular white matter in participant 3 (0.027 intact proviruses/million cells), in whom analysis was limited to brain tissues. No intact proviruses were detected in the CNS tissues of study participant 2, despite analyzing 66.29 million cells. Together, these results indicate that intact HIV-1 proviruses are preferentially detected in lymphoid and gastrointestinal (GI) tissues. The frequency of intact proviruses in the CNS is comparatively low; however, this study is the first one to document the presence of genome-intact proviral sequences in CNS tissues using near full-length proviral sequencing.

Defective proviral sequences were detected in all analyzed tissue samples except for the thalamus from participant 1 (**Figure 2C**). In participant 1, the frequency of defective proviral sequences was highest in lymph nodes (5.7 defective proviruses/million cells), followed by colon, spleen, and kidney. In participant 2, the frequency of defective proviral sequences was highest in lymph nodes (13.0 defective proviruses/million cells), followed by prostate, colon, and spleen. Notably, the prostate had a very high frequency of virally-infected cells. Taken together, these results demonstrate the highest frequencies of defective proviruses in lymph nodes, in the colon and in the prostate. CNS tissues contained relatively low frequencies of proviral sequences, compared to other tissue sites; however, defective proviruses were isolated in all but one of the analyzed 13 different CNS samples.

Phylogenetic associations and clonality

A series of prior studies demonstrated large sequence-identical clones of intact and defective proviruses in the peripheral blood of ART-treated study participants. In our subsequent analysis, we studied the dissemination of clonal proviral sequences across different tissues. In participant 1, a total of 24 intact proviral sequences were detected. Two large clones of intact proviruses were observed, one of which included sequences detected in kidney, lymph node, and spleen samples (**Figure 3A-B**). The other clone involved intact proviral sequences from basal ganglia and lymph node tissues. Among 218 defective proviruses detected in participant 1, 14 clones were observed across multiple tissues (**Figure 3C**). In participant 2, among 22 intact proviral sequences, only

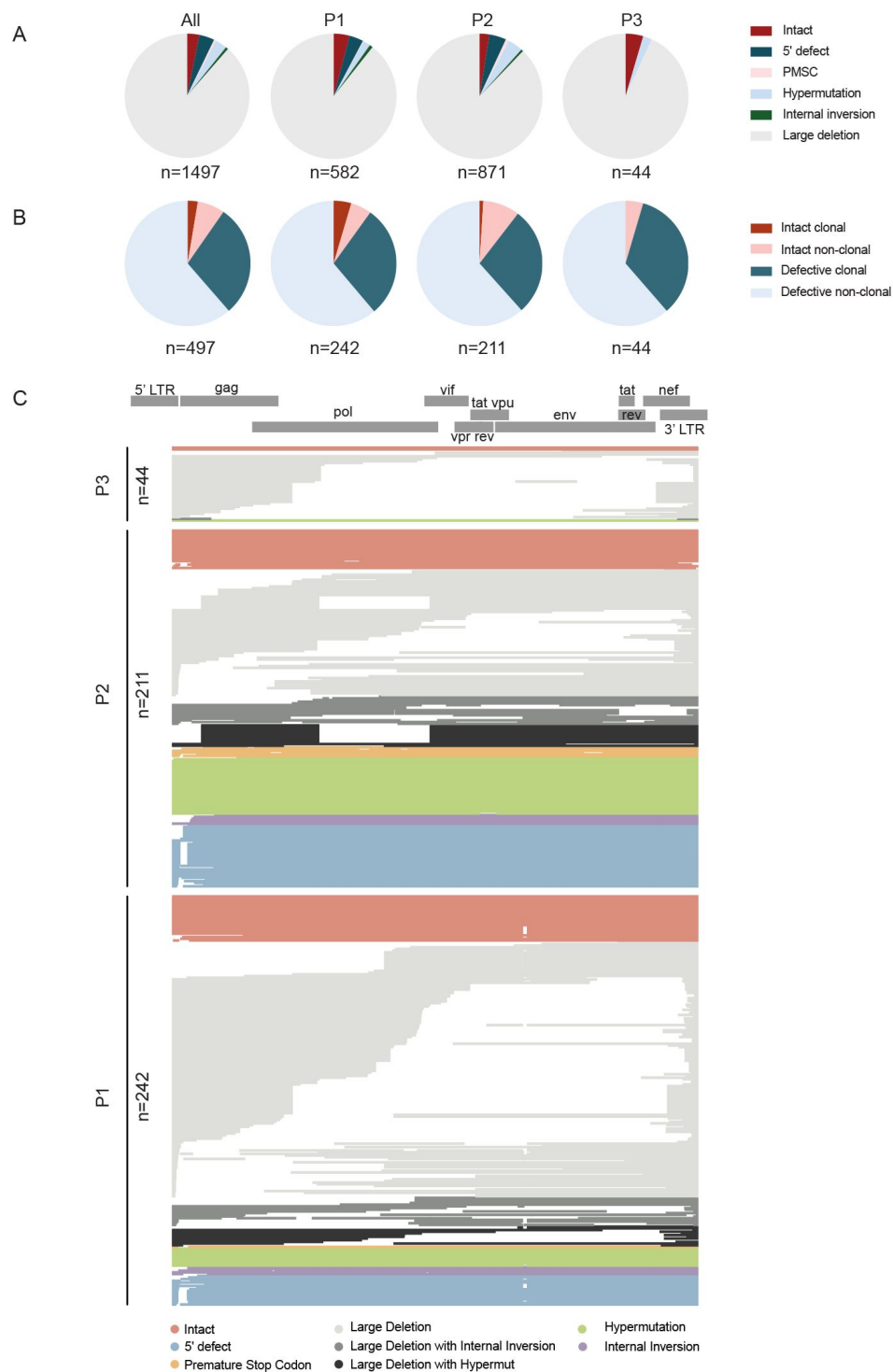


Figure 1.

Proviral sequence classification in all analyzed HIV-1-infected cells from 3 study participants.

(A) Pie charts reflecting proportions of proviruses classified as intact or defective. All proviruses identified by single-genome, near-full-length, next-generation sequencing and by counting amplification products in agarose gel electrophoresis were included. The total number of individual sequences included is listed below each pie chart. (B) Pie charts indicating proportions of intact and defective proviruses detected once (classified as non-clonal) and detected multiple times (classified as clonal). The total number of proviral sequences identified by single-genome, near-full-length, next-generation sequencing is listed below each pie chart. (C) Virograms summarizing individual HIV-1 proviral sequences aligned to the HXB2 reference genome from each participant; color coding reflects the classification of proviral sequences.

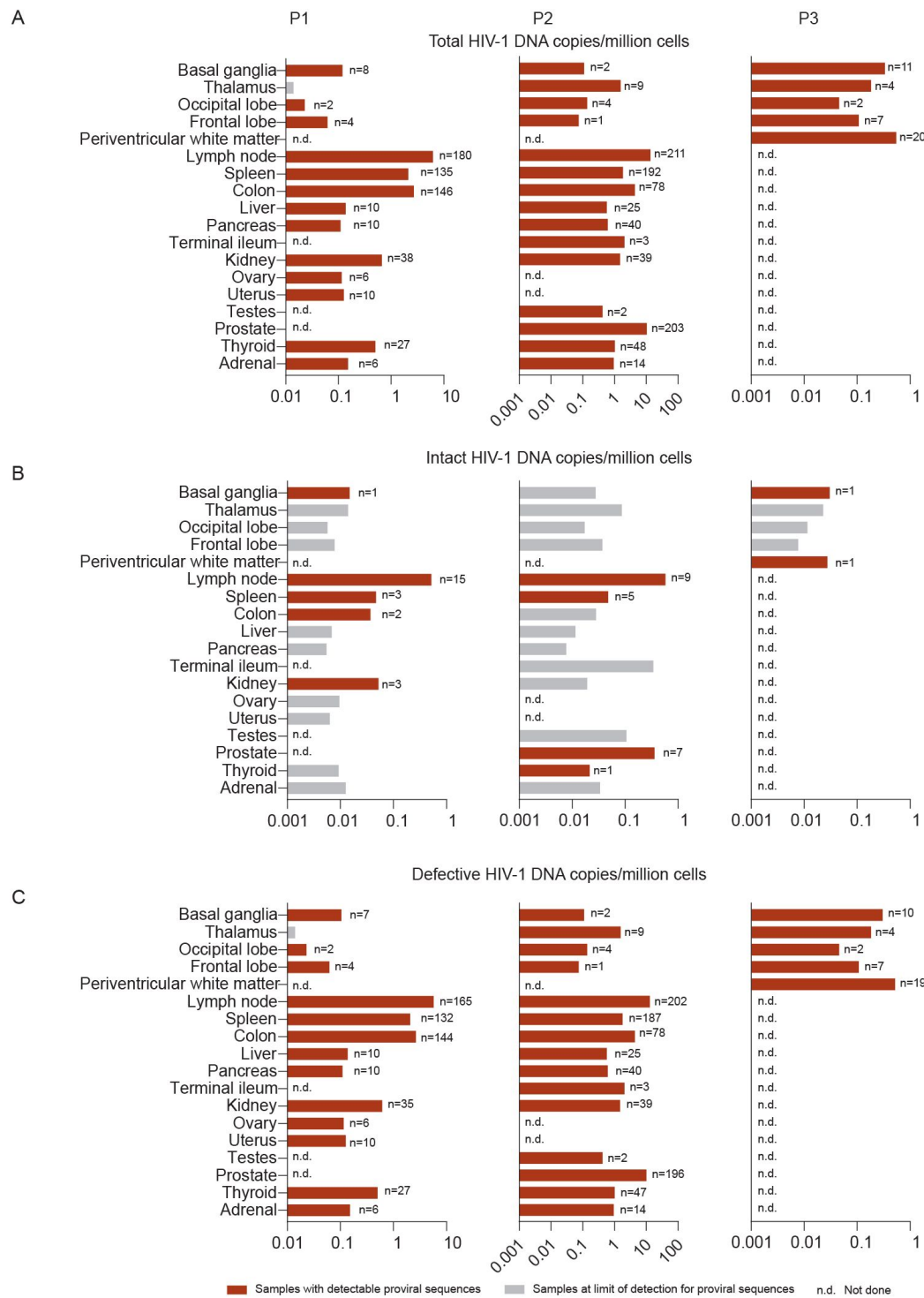


Figure 2.

Distribution of total, intact, and defective HIV-1 proviruses in individual tissue compartments.

Bar diagrams reflect relative frequencies of total (A), intact (B), and defective (C) proviruses in all analyzed tissues in study subjects 1-3. The total number of individual proviral sequences determined by single-genome, near-full-length, next-generation sequencing and by counting amplification products in agarose gel electrophoresis from each tissue site of each participant is listed aside each bar. The red bars reflect samples with detectable proviral sequences; grey bars reflect samples at limit of detection for proviral sequences, calculated as 0.5 (single genome near-full-length PCR) copies per maximum number of cells tested without target identification (see Materials and Methods for details). N.d. indicates that the samples were not available from the indicated tissue sites.

one clone with two member sequences was identified; both of these clonal sequences were located in the prostate (**Figure 3A-B**). Fourteen clones of defective proviruses were observed across multiple tissues in participant 2 (**Figure 3C**).

Clonal proviral sequences in CNS tissues were detected in all three study participants. In study participant 1, members of a clone of defective proviruses were observed in the basal ganglia and the frontal lobe (**Figure 4A-B**). In participant 2, two clones of defective proviruses were noted in the thalamus, one clone was detected in cells from the basal ganglia, and members of a fourth clone were detected in the thalamus and the occipital lobe (**Figure 4C-D**), demonstrating rather extensive evidence for clonal proliferation of virally-infected cells in the brain. Two of the 44 proviral sequences detected in the central nervous system of participant 3 met the criteria for genome-intactness; one of those was isolated from the basal ganglia and one from periventricular white matter. Defective proviruses, most of them harboring large deletions, were detected in all 5 CNS tissues from participant 3 (**Figure 4E-F**). One large clone involving 9 defective proviral sequences in participant 3 was broadly distributed across different brain tissues, encompassing sequences in the occipital lobe, basal ganglia, thalamus, and periventricular white matter. The other clone of defective proviruses was only detected in the periventricular white matter. Again, these data suggest that clonal proliferation is a rather common feature of HIV-1 reservoir cells in the CNS.

Viral tropism and immune selection footprints

Viral tropism was evaluated based on the env V3 region of the proviruses using the Geno2pheno algorithm. Notably, all proviral sequences containing the env V3 region of participant 1 (n=130, 53.7%) and 3 (n=6, 13.6%) were predicted to be CCR5-tropic (**Figure 5A**). In participant 2, 63.5% of proviral sequences (n=134) were predicted to be non-CCR5-tropic, while 17.1% (n=36) were classified as CCR5-tropic; the remaining 19.4% (n=41) were classified as undetermined due to the lack of env V3 regions in this study participant (**Figure 5A**). Notably, among 134 non-CCR5 proviruses from study participant 2, half of them (n=68, 50.7%) were isolated from the prostate, followed by the spleen (n=38, 28.4%), lymph node (n=20, 14.9%) and thyroid gland (n=4, 3.0%) (**Figure 5B-C**). Among all proviruses (n=16) from CNS tissues of participant 2, only 1 provirus with a large deletion, isolated from the occipital lobe, was non-CCR5; the tropism of other proviruses from the CNS was undetermined due to the lack of env V3 regions. As an additional analysis step, we evaluated footprints of immune selection pressure and mutations resulting in resistance to antiretroviral agents in intact proviral sequences from our study subjects. We noted that the frequency of viral amino acid residues associated with resistance to broadly-neutralizing antibodies did not notably differ among sequences isolated from different tissue compartments (**Figure S2**). We did not observe sequence variations consistent with escape from antiretroviral agents in any of the intact proviral sequences analyzed here.

Discussion

The lifelong persistence of viral reservoir cells makes HIV-1 infection an incurable disease that necessitates indefinite antiretroviral suppression therapy. However, the location of HIV-1 viral reservoir cells across different tissues has been difficult to assess in the past, due to the limited availability of tissue samples. In our study, we used single-genome near full-length proviral sequencing to evaluate the distribution of HIV-1 reservoir cells in multiple anatomical compartments from autopsy samples of three individuals living with HIV-1 and receiving antiretroviral therapy until the time of their decease. Consistent with previous studies (23, 24), intact proviruses were readily detected in lymph nodes; moreover, we detected relatively high frequencies of intact proviruses in the colon, likely reflecting viral infection of CD4 T cells residing in gut-associated lymphoid tissues (GALT). Of note, our study is the first one to identify

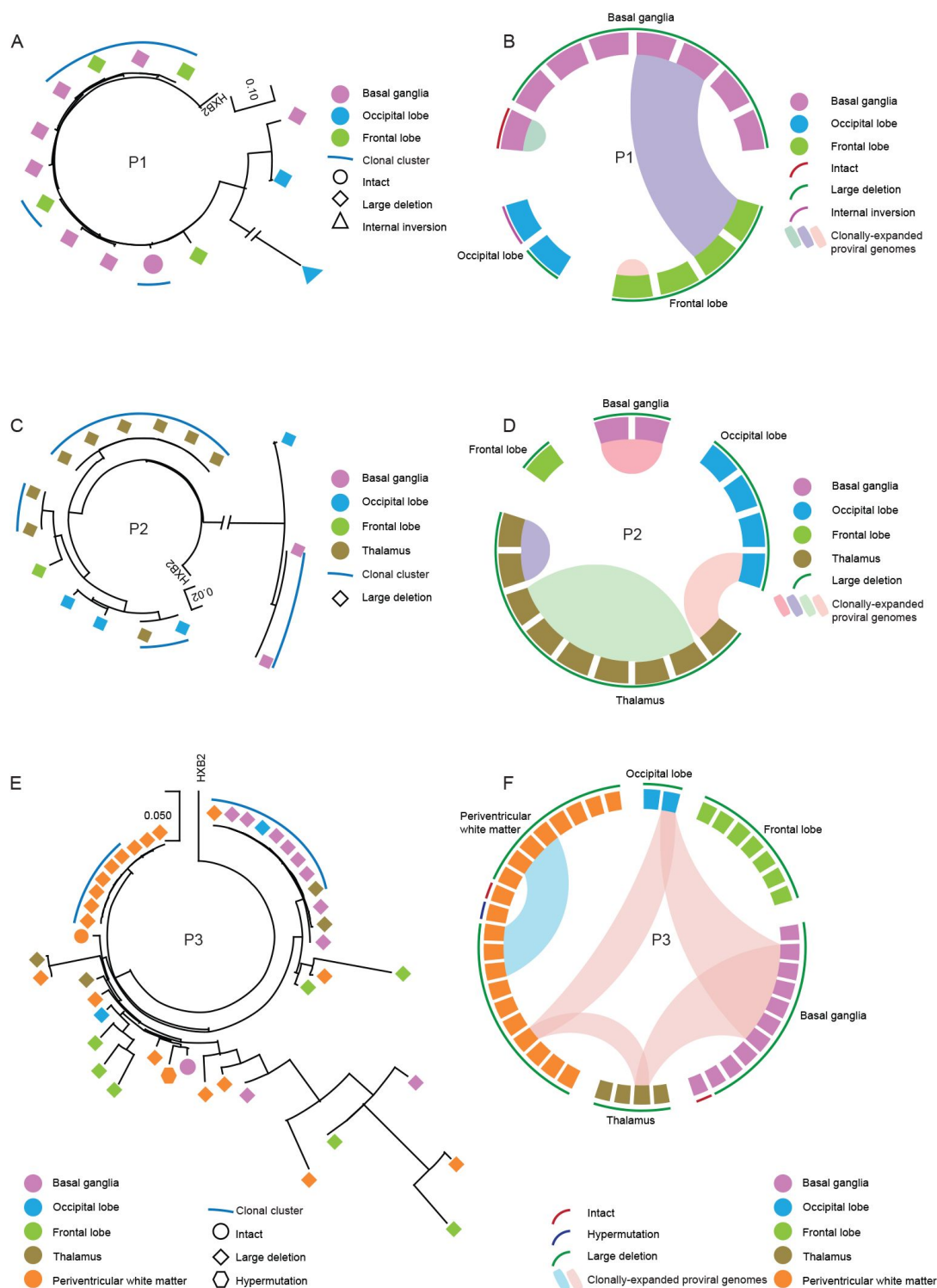


Figure 4.

Dissemination of HIV-1-infected cells across CNS tissues.

(A, C, and E) Circular maximum likelihood phylogenetic trees of all proviral sequences derived from CNS tissues of the 3 study participants (A, participant 1; C, participant 2; E, participant 3). Color coding reflects tissue origins. Clonal sequences, defined by complete sequence identity, are indicated by blue arches. (B, D, and F) Circos plot reflecting the clonality of all proviral sequences isolated from CNS tissues of 3 participants (B, participant1; D, participant 2; F, participant 3). Each symbol reflects one provirus. Clonal sequences, defined by complete sequence identity, are highlighted. Color-coded arches around the plots indicate types of proviral sequences.

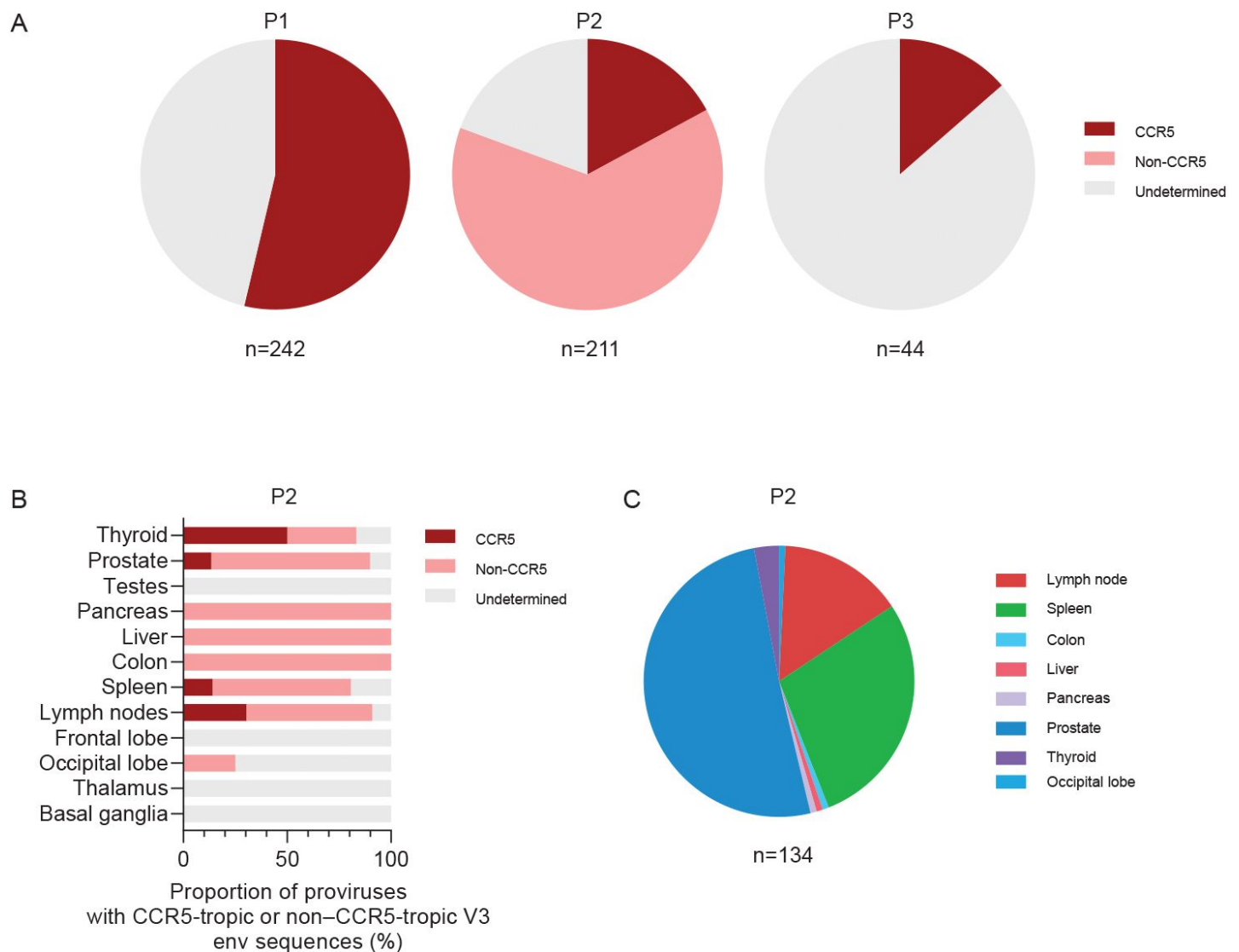


Figure 5.

HIV-1 tropism analysis of proviral sequences.

(A) Pie charts indicating the proportions of all proviruses from each participant with CCR5-tropic or non-CCR5-tropic V3 envelope sequences are shown. The total number of proviral sequences included in this analysis is listed below each pie chart. (B) The proportions of proviruses with CCR5-tropic or non-CCR5-tropic V3 env sequences in each tissue from participant 2 are shown. (C) Pie charts indicating the proportions of all non-CCR5-tropic proviruses from participant 2 are shown. Color coding reflects tissue origins. The total number of analyzed proviral sequences is listed below the pie chart. HIV-1 tropism was computationally inferred using Geno2pheno (<https://coreceptor.geno2pheno.org/>).

near full-length proviral sequences from the central nervous system in two study participants, supporting the hypothesis that CNS cells can serve as reservoirs for long-term HIV-1 persistence despite antiretroviral therapy. Importantly, we noted that large clones of virally infected cells were broadly disseminated across multiple tissues, and, in selected cases, involved cells from the CNS; this suggests that HIV-1 reservoir cells seeded to the brain via hematogenous spread can proliferate in the local tissue microenvironment of the central nervous system. Together, our work suggests that HIV-1 reservoir cells harboring intact proviruses are broadly distributed across multiple anatomical locations, involving lymphoid tissues, gastrointestinal tissues, genitourinary tissues, and, importantly, central nervous system tissues.

HIV-1 can invade the central nervous system (CNS) within days after infection, as demonstrated in animal (25) and in human studies (26). Most likely, infection of CNS cells occurs as a result of transmigration of infected CD4 T cells and macrophages across the blood-brain-barrier (BBB) (27, 28), a process that may be facilitated by the increased permeability of the BBB during the initial, highly-replicative stage of HIV-1 infection. Infected cells that successfully entered the CNS are probably short-lived and unlikely to persist long-term. However, invading infected cells can transmit the virus to resident CNS cells via effective cell-to-cell transmission (28). At least three different CNS cell types seem to be susceptible to HIV-1 infection: perivascular macrophages, microglial cells, and astrocytes, although the role of the latter as HIV-1 target cells is more controversial (29, 30). Yet, due to the difficulties in accessing brain tissues for analytic purposes, the role of the central nervous system in HIV-1 persistence during antiretroviral therapy remained largely unknown for a long time. Using the intact proviral DNA assay (IPDA), a ddPCR-based technique allowing to identify proviruses with high probability of being genome-intact, previous investigators identified intact proviruses in 6 out of 9 ART-treated persons (17), although the precise proviral DNA sequence, and their possible clonality was not assessed with this technology. In our study, a total of 13 CNS tissue samples from three study subjects were analyzed, including specimens from the basal ganglia, thalamus, occipital lobe, frontal lobe, and periventricular white matter. In two study persons (participants 1 and 3), intact proviral sequences were detected in basal ganglia, suggesting that HIV-1 may preferentially persist in this anatomical compartment in the CNS; an additional intact provirus was detected in periventricular white matter. Notably, the intact provirus from basal ganglia in one of our study persons (participant 1) was clonal with 4 intact proviruses from the lymph node, indicating, to our knowledge for the first time, that CNS tissue can be involved in the multi-compartment dissemination of large clones of HIV-1 proviruses in ART-treated persons. Moreover, in multiple instances, we observed clones of HIV-1-infected cells that were distributed across different autologous CNS tissues, specifically in study subject 3, suggesting local spread of virally infected cells through clonal proliferation within the immune microenvironment of the CNS.

Our study did not allow to determine which cell types were infected by HIV-1 and responsible for clonal expansion of viral reservoir cells in the CNS, however, we suspect that infected microglia are most likely to be involved. Microglial cells originate from erythromyeloid progenitors in the yolk sac and colonize the developing CNS during embryogenesis (31); these cells act as the main innate immune cell population of the CNS. Due to their long half-life (typically several years), their ability to divide and self-renew, and their high cell-intrinsic susceptibility to HIV-1 (32), these cells may represent a primary cellular site for long-term HIV-1 persistence in the CNS. In particular, self-renewal through homeostatic proliferation in microglia (33) may support HIV-1 persistence through clonal expansion. A recent study indeed identified HIV-1 DNA and RNA in microglia cells from autopsies from ART-treated PWH who did not have specific (HIV or non-HIV associated) CNS pathology (34). Evidence for HIV persistence in CD68+ myeloid cells, most likely microglia cells, was also described by previous investigators (17).

In our study, non-CCR5 (likely CXCR4-tropic) proviruses were exclusively detected in participant 2. Compared to the other two participants who began ART shortly after HIV-1 diagnosis, participant 2 was diagnosed with HIV-1 25 years prior to starting antiretroviral therapy and died 10 months

after ART commencement; therefore, viral CXCR4 tropism most likely resulted from “coreceptor switch” frequently occurring during advanced stages of immune deficiency (35 [↗](#)). Whether the preferential persistence of non-CCR5-tropic viruses in study subject 2 was associated with our inability to detect intact proviruses in the CNS in this person is unclear; however, prior studies suggested that CCR5 can act as the principal co-receptor for HIV-1 isolates in the brain (36 [↗](#), 37 [↗](#)). Moreover, rebound viremia in cerebrospinal fluid after ART interruption is mostly fueled by CCR5-tropic virus (38 [↗](#)), further supporting the assumption that R5-tropic viruses are better adjusted to persist in the brain. Another notable finding in our study was the high number of HIV-1 proviruses isolated from the prostate, which had the second highest proviral frequency among all analyzed tissues in study participant 2, second only to lymph node samples. Other studies also reported that the prostate can represent a tissue reservoir for HIV-1 (39 [↗](#)). We were unable to identify the precise cell type harboring HIV-1 in the prostate, but it is possible that myeloid cells may harbor HIV-1 in this location. A prior study indeed demonstrated that intact, replication-competent HIV-1 can persist in myeloid cells from the urethra, located in immediate anatomical proximity to the prostate (40 [↗](#)).

Our study has several limitations. Importantly, this work includes only 3 participants, and brain tissues were the only samples available from participant 3. Moreover, due to limited tissue sizes, very few cells were assayed from the terminal ileum and testes; prior studies suggested high HIV-1 DNA levels in these 2 tissues during suppressive antiretroviral therapy (41 [↗](#), 42 [↗](#)). Another limitation was that peripheral blood samples were not available from the 3 participants, which made it impossible to study phylogenetic associations between tissue reservoirs of HIV-1 relative to viral species circulating in peripheral blood. Moreover, we cannot fully exclude contamination of tissue samples with cells from peripheral blood. However, after autopsy, the tissue samples were washed thoroughly with PBS to eliminate blood as much as possible. In addition, we failed to detect intact proviral sequences from over 100 million liver cells of 2 participants, despite the fact that the liver contains about 13% of all human blood supply at any given time point, arguing against contamination of tissues with blood cells.

In sum, this study provides a deep analysis of tissue reservoirs for HIV-1 that includes a detailed assessment of HIV-1 sequences in CNS tissues. Our work supports the persistence of genome-intact HIV-1 in many tissues, including CNS tissues, emphasizing the difficulties in finding strategies to effectively eliminate HIV-1 from the human body in clinical settings.

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Author contributions

Sample collection and preservation: SIC, IHS, SM

HIV-1 sequencing assays: WS, YR

Data analysis, interpretation, presentation: WS, CG, XL, XGY, ML

Conflicts of Interest

The authors declare that conflicts of interest do not exist.

Supplemental Material

Supplemental Figure 1: (A) Clinical and demographical data of study participants. (B) Cell numbers analyzed from each tissue of each study participant.

Supplemental Figure 2: (A-F) Numbers of broadly neutralizing antibody (bnAb) sensitivity (A, C, and E) and resistance (B, D, and F) signature sites in intact proviral sequences from indicated tissues from each participant are shown. Each dot represents one intact provirus. Amino acid residues associated with susceptibility or resistance to bnAbs were inferred based on the study by Bricault (21 [↗](#)).

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Reviewer #1 (Public Review):

Despite durable viral suppression by antiretroviral therapy (ART), HIV-1 persists in cellular reservoirs in vivo. The viral reservoir in circulating memory T cells has been well characterized, in part due to the ability to safely obtain blood via peripheral phlebotomy from people living with HIV-1 infection (PWH). Tissue reservoirs in PWH are more difficult to

sample and are less well understood. Sun and colleagues describe isolation and genetic characterization of HIV-1 reservoirs from a variety of tissues including the central nervous system (CNS) obtained from three recently deceased individuals at autopsy. They identified clonally expanded proviruses in the CNS in all three individuals.

Strengths of the work include the study of human tissues that are under-studied and difficult to access, and the sophisticated near-full length sequencing technique that allows for inferences about genetic intactness and clonality of proviruses. The small sample size ($n=3$) is a drawback. Furthermore, two individuals were on ART for just one year at the time of autopsy and had T cells compatible with AIDS, and one of these individuals had a low-level detectable viral load (Figure S1). This makes generalizability of these results to PWH who have been on ART for years or decades and have achieved durable viral suppression and immune reconstitution difficult.

While anatomic tissue compartment and CNS region accompany these PCR results, it is unclear which cell types these viruses persist in. As the authors point out, it is possible that these reservoir cells might have been infiltrating T cells from blood present at the time of autopsy tissue sampling. Cell type identification would greatly enhance the impact of this work. Several other groups have undergone similar studies (with similar results) using autopsy samples (links below). These studies included more individuals, but did not make use of the near-full length sequencing described here. In particular, the Last Gift cohort, based at UCSD and led by Sara Gianella and Davey Smith, has established protocols for tissue sampling during autopsy performed soon after death.

<https://pubmed.ncbi.nlm.nih.gov/35867351/>

<https://pubmed.ncbi.nlm.nih.gov/37184401/>

Overall, this small, thoughtful study contributes to our understanding of the tissue distribution of persistent HIV-1, and informs the ongoing search for viral eradication.

Reviewer #2 (Public Review):

The manuscript by Sun et al. applies the powerful technology of profiling viral DNA sequences in numerous anatomical sites in autopsy samples from participants who maintained their antiviral therapy up to the time of death. The sequencing is of high quality in using end-point dilution PCR to generate individual viral genomes. There is a thoughtful discussion, although there are points that we disagree with. This is an important data set that increases the scope of how the field thinks about the latent reservoir with a new look at the potential of a reservoir within the CNS.

1. The participants are very different in their exposure to HIV replication and disease progression. Participant 1 appears to have been on ART for most of the time after diagnosis of infection (16 years) and died with a high CD4 T cell count. The other two participants had only one year on ART and died with relatively low CD4 T cell counts (under 200). This could lead to differences in the nature of the reservoir. In this regard, the amount of DNA per million cells appears to be about 10-fold lower across the compartments sampled for participant 1. Also, one might expect fewer intact proviruses surviving after 16 years on ART compared to only 1 year on ART. The depth of sampling may be too limited and the number of participants too few to assess if these differences are features of these participants because of their different exposures to HIV replication. On the positive side, finding similarities across these big differences in participant profiles does reinforce the generalizability of the observations.

2. The following analysis will be limited by sampling depth but where possible it would be interesting to compare the ratio of intact to defective DNA. A sanctuary might allow greater persistence of cells with intact viral DNA even without viral replication (i.e. reduced immune surveillance). Detecting one or two intact proviruses in a tissue sample does not lend itself to

a level of precision to address this question, but statistical tests could be applied to infer when there is sampling of 5 or more intact proviruses to determine if their frequency as a ratio of total DNA in different anatomical sites is similar or different. This would allow adjustment for the different amount of viral DNA in different compartments while addressing the question of the frequency of intact versus defective proviruses. One complication in this analysis is if there was clonal expansion of a cell with an intact genome which would represent a fortuitous over-representation intact genomes in that compartment.

3. The key point of this work is that the participants were on therapy up to the time of death ("enforcing" viral latency). The predominance of defective genomes is consistent with this assumption. Is there data from untreated infections to compare to as a signature of whether the viral DNA population was under selective pressure from therapy or not? Presumably untreated infections contain more intact DNA relative to total DNA. This would represent independent evidence that therapy was in place.

4. There are several points in Figure 5 to raise about V3 loop sequences. The analysis includes a large number of "undetermined" sequences that did not have a V3 loop sequence to evaluate. We would argue it is a fair assumption that the deleted proviruses have the same distribution of X4 and R5 sequences as the ones that have a V3 sequence to evaluate. In this view it would be possible to exclude the sequences for which there is no data and just look at the ratio of X4 and R5 in the different compartments, specifically does this ratio change in a statistically significant way in different compartments? The authors use "CCR5 and non-CCR5" as the two entry phenotypes. The evidence is pretty strong that the "other" coreceptor the virus routinely uses is CXCR4, and G2P is providing the FPR for X4 viruses. Perhaps the authors are trying to create some space for other coreceptors on microglia, but we are pretty sure what they are measuring is X4 viruses, especially in this late disease state of participant 2. Finally, we have previously observed that the G2P FPR score of <2 is a strong indicator of being X4, FPR scores between 2 and 10 have a 50% chance of being X4, and FPR scores above 10 are reliably R5 (PMID27226378). In addition, we observed that X4 viruses form distinct phylogenetic lineages. The authors might consider these features of X4 viruses in the evaluation of their sequences. Specifically, it would be helpful to incorporate the FPR scores of the reported X4 viruses.

5. We have puzzled over the many reports of different cell types in the CNS being infected. When we examined these cells types (both as primary cells and as iPSC-derived cells), all cells could be infected with a version of HIV that had the promiscuous VSV-G protein on the virus surface as a pseudotype. However, only macrophages and microglia could be infected using the HIV Env protein, and then only if it was the M-tropic version and not the T-tropic version (PMID35975998). RNAseq analysis was consistent with this biological readout in that only macrophages and microglia expressed CD4, neurons and astrocytes do not. From the virology point of view, astrocytes are no more infectable than neurons.

6. The brain gets exposed to virus from the earliest stages of infection but this is not synonymous with viral replication. Most of the time there is virus in the CSF but it is present at 1-10% of the level of viral load in the blood and phylogenetically it looks like the virus in the blood, most consistent with trafficking T cells, some of which are infected (PMID25811757). The fact that the virus in the blood is almost always T cell-tropic in needing a high density of CD4 for entry makes it unlikely that monocytes are infected (with their low density of CD4) and thus are not the source of virus found in the CNS. It seems much more likely that infected T cells are the "Trojan Horse" carrying virus into the CNS.

7. While all participants were taking antiretroviral therapy at the time of their death, they were not all suppressed when the tissues were collected. The authors are careful not to mention "suppressive ART" in the text, which is appreciated. However, the title should be changed to also reflect this fact.